Data Analytics for Immersive Environments - Individual Project

Interactive RDBMS data with Quarto & Shiny

Muhammad Adil Raja

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This assignment is designed for a postgraduate program with a focus on Data Analytics for Game & Extended Reality and Computer Animation. It encompasses five distinct parts, each aimed at assessing and enhancing students' proficiency in different aspects of data analytics as applied to game development and animation.

Table of Contents

Learning Outcomes

To practice the following:

- Writing and executing complex SQL queries to retrieve and manipulating data.
- Understanding and applying SQL concepts in a database context.
- Analysing and interpreting data from a SQL database effectively.
- Developing and interpreting linear regression models in the context of game development data.

- Understanding the relationship between different variables and how they impact each
- other.
- Evaluating the performance of a linear regression model and understanding its limitations
- and applicability.
- Utilising Quarto to create interactive Shiny web applications.
- Designing user-friendly interfaces for data interaction and visualisation.
- Implementing data visualisation techniques to effectively communicate data insights.
- Applying data analytics techniques to real-world scenarios in game development.
- Analysing and interpreting data specific to game development projects or products.
- Demonstrating the ability to solve problems by applying analytical skills in a structured manner.

Functional Requirements

Part A – SQL (20 marks)

You will be provided with an SQLite database file (via Moodle before **5PM 15th December 2025**) containing information about **interactive cinematic virtual production**. This database includes tables for projects, assets, developers, timelines, and customers. The structure of each table is shown.

Productions

Represents each film, commercial, music video, or short that uses virtual production.

Column	Type	Description
ProductionID	Integer (PK)	Unique ID
Title	Text	Production name
Genre	Text	Sci-Fi, Drama, Commercial…
StartDate	Date	Pre-production start date
EndDate	Date	Completion date
Budget	Decimal	Total budget
Status	Text	In Planning / Shooting / Post-production
ClientID	Integer (FK)	Link to Clients table

Clients

Represents agencies, brands, studios commissioning productions.

Column	Type
ClientID	Integer (PK)
ClientName	Text
City	Text
Country	Text

Environments

Unreal Engine worlds displayed on LED volume or greenscreen composites.

Column	Type
EnvironmentID	Integer (PK)
ProductionID	Integer (FK)
EnvironmentName	Text
EnvironmentType	Text (Urban / Forest / Sci-Fi Lab / Desert)
EngineVersion	Text (UE 5.1, UE 5.3…)
CreationDate	Date

Assets

Individual assets used within Unreal Engine environments.

Column	Type
AssetID	Integer (PK)
EnvironmentID	Integer (FK)
AssetName	Text
AssetType	Text (3D Model / Animation / Texture / Blueprint / HDRI)
FileSizeMB	Decimal
CreatedDate	Date

CrewMembers

Everyone involved in virtual production.

Column	Type
CrewID	Integer (PK)
Name	Text
Role	Text (Unreal Operator, VCam Specialist, DoP, MoCap Tech)
ExperienceYears	Integer

ProductionCrew

Links developers to the projects they are working on and shows which developer worked on which project and their specific role in that project.

Column	Type
ProductionID	Integer (FK)
CrewID	Integer (FK)
Responsibility	Text (Lighting, LED Control, Scene Optimization…)

ShootingSesion

Actual shooting days where LED or MoCap is used.

Column	Type
SessionID	Integer (PK)
ProductionID	Integer (FK)
EnvironmentID	Integer (FK, optional)
Date	Date
DurationHours	Decimal
Notes	Text

Vcam Takes

Virtual camera takes recorded during sessions.

Column	Type
TakeID	Integer (PK)
SessionID	Integer (FK)
TakeNumber	Integer
CameraOperatorID	Integer (FK to CrewMembers)
DurationSeconds	Integer
TrackingQuality	Text (Excellent / Good / Poor)

Equipment

Tracking systems, LED walls, camera rigs, mocap suits.

Column	Type
EquipmentID	Integer (PK)
EquipmentName	Text
Type	Text (CameraRig, MotionCapture, LEDWall, MarkerlessTracking…)
Manufacturer	Text
PurchaseYear	Integer

Session Equipment

Which equipment was used in which shooting session.

Column	Type
SessionID	Integer (FK)
EquipmentID	Integer (FK)
Purpose	Text (LED Rendering, Camera Tracking, Body MoCap)

First, you are required to write three core SQL queries to perform the following tasks:

1. List the total budget allocated for productions in each country (via Clients + Productions), along with the count of projects per country. Display sorted by the total budget in descending order.
2. List the average development time for productions, categorized by the number of assets used.
3. List the top three crew members based on the number of successful productions they've been involved in. Display the results.

Next, you are also required to demonstrate the following three general SQL concepts using no fewer than three distinct SQL statements:

1. SELECT with LIKE and OR
2. SELECT with DISTINCT and ORDER BY
3. Subquery with SELECT

Display the results in formatted table(s) in a Quarto Notebook file inside a Quarto Website. You should add pagination to the tables if the number of rows is larger than 8-10 (see Recommended Reading on creating tables).

Part B - Linear Regression (20 Marks)

Utilise the provided dataset, which contains details about production budgets, timelines, team sizes, and success rates and apply linear regression analysis to understand trends or patterns in the cinematic life cycle.

You are required to perform the following tasks:

1. Model: Perform linear regression to predict the success rate of a production based on its budget and crew size and present the data in an appropriate plot.

2. Interpret: Interpret the model coefficients and discuss what insights they provide about cinematic production.

3. Discuss: Comment on the reliability of the the linear regression model and any outliers present in the data.

Output the results in a Quarto Notebook file inside the **Quarto Website** from Part A.

Part C - Interactive Shiny Application using Quarto Dashboard (30 Marks)

Create an interactive Shiny application using Quarto Dashboards to display and interact with data from the database. You are required develop an interactive web application containing

two pages/tabs which allow the following:

1. **Data View:** The first page/tab should allow the user to view the contents of **any four tables** in the database with no filtering or queries executed. You may need to consider using a more advanced table library (e.g. kable + kableExtra, reactable, DT) to add pagination to larger tables.

2. **Plot View:** The second page/tab should allow users to visualize data through 2-3 appropriate plots and **interact** with **each** plot. Typically, plots could include data related to production timelines, budget, asset types, etc.

The second page/tab must support user interactions via input fields (e.g., textfield, dropdown, slider, checkbox) to allow interactions such as: * Selecting crew members to view more detailed data on the productions they participated in. * Specifying a completion date range for productions. You must include a minimum of **two fields** for interaction for **each** plot.

Output the results in a **Quarto Website**. The web site does not need to be published to a remote server and may be run from your machine. This will be a second, and separate project from the Quarto Website produced for Parts A & B.

Part D - Technology Exploration (10 Marks)

Finally, you will be awarded a maximum of 10 marks for creative use of R packages which have not been covered in class. For example, this could be a package for rendering more visually appealing tables (i.e. reactable), animating plots (i.e. gganimate), or rendering the geospatial data from the database (i.e. customer city and customer country) using additional R packages (i.e. ggplot2 and ggmap).

Inside the **Quarto Website** from Parts A & B, you should include a short paragraph in a separate page, entitled Technology Exploration.qmd, which technologies you have explored in Parts A, B & C.

Part E - Quality & Conclusions (20 Marks)

You are required to present **two** separate project repos:

1. **Quarto Website** which brings together Parts A, B, D.
2. **Quarto Website** which hosts the Quarto Dashboard for Part C.

You will be assessed in terms of adherence to best practices in coding and data visualization and the quality of your conclusions. The assessment criteria for this component are listed below:

Document Formatting (4 Marks)

- Overall structure and organization of the report.
- Effective use of headings, subheadings, and bullet points.
- Consistency in font, spacing, and margins.
- Proper pagination and table of contents (if applicable).

Suitability of Tables in Part A (4 Marks)

- Clarity and readability of tables.
- Appropriate labeling for tables.
- Consistent and legible formatting in tables and figures.

Suitability of Plots in Part B (4 Marks)

- Relevance and effectiveness of chosen plots to represent the data.
- Ability to convey the intended message or findings clearly.
- Innovative and insightful use of data visualization techniques.

Adherence to Best Practices in Coding and Data Visualization (5 Marks)

- Code clarity, including commenting and organization.

- Use of efficient and appropriate coding methods.
- Adherence to data visualization best practices, including color choice, scale, and avoiding misleading representations.

Conclusion and Reflection (4 Marks)

- Inclusion of a 1-2 paragraph Conclusion section.
- Depth of reflection on what was learned from each part of the assignment.
- Insightful discussion on how the skills and knowledge gained apply to game development and computer animation.

Version Control Requirements

You must use a recognized online code repository (e.g., GitHub) and make regular well-named commits to your private repositories. Two links to your repos must be included in a README as part of the final submission and you must add your lecturer as a developer to the repositories. The repository must be named 2025_DAIE_ICA_1_StudentInitials and 2025_DAIE_ICA_2_StudentInitials.

Your grade for this component will depend directly upon the **regularity** of your commits. A development project of this size should consist of a minimum of **5+ distinct commit messages** per repository, spread over the lifetime of the development. Committing all your code in one commit, before the deadline, will be interpreted negatively.

Any submission made which does not include a repository link will not be graded.

Submission Requirements

1. In the README of your repositories you must include the **main file name** to be opened for each of the two projects (e.g. index.qmd for Part C Application) for each requirement. I will not filter through your submission looking for the correct file for each part. I will interpret the absence of this file as a failure to properly communicate your work which will negatively affect your grade for Part E.
2. A link to your R source code repositories (i.e. Quarto Website, Shiny Application). Ensure that **no** changes are made to the repositories following the submission deadline. You should create a separate fork for this submission and leave it unchanged after the deadline. Ask your lecturer for details on fork creation.
3. The assignment must be entirely the work of each student. Students are **not** permitted to share any pseudo-code or source code from their solution with any other group in the class.
4. Students may **not** distribute the source code of their solution to any other student, in any format (i.e., electronic, verbal, or hardcopy transmission).
5. Plagiarised assignments will receive a mark of **zero**. This also applies to the individual/group allowing their work to be plagiarised. Any plagiarism will be reported to the Head of Department and a report will be added to your permanent academic record.

6. Late assignments will **only** be accepted if accompanied by the appropriate medical note. This documentation must be received within 10 working days of the project deadline. The Institute standard penalties for late submission will apply.
7. Each student **must** complete and **sign** a single assignment cover sheet. Please submit the signed cover sheet before 5 pm on the Friday of the week of the deadline.

Support Material

- GitHub - Quarto Website Demo
- GitHub - Shiny App with Quarto Dashboard
- Screencast - Setting up a Quarto Dashboard
- GitHub - Connecting to an SQLite DB

Recommended Reading

- Recommended Text - OpenIntro Statistics Videos & Slides
- SQL Tutorial
- A Comprehensive Introduction to Working with Databases using R
- Simple Linear Regression | An Easy Introduction & Examples
- Linear regression using R programming

Quarto Markdown & Shiny Related Resources

- YouTube - Hello, Quarto: A World of Possibilities (for Reproducible Publishing)
- Quarto - Using R
- Quarto Dashboards | Charles Teague | Posit
- Dashboards with Shiny for R

Plot & Table Related Resources

- The R Graph Gallery
- Get started with esquisse
- Analyze Data quickly with Esquisse
- Add Regression Line to ggplot2 Plot in R
- Fitting and visualizing linear regression models with the ggplot2 R package (CC237)
- Visualize your data using ggplot. R programming is the best platform for creating plots and graphs.
- Drag-and-drop ggplot2 graphs with the Esquisse library
- ggplot for plots and graphs. An introduction to data visualization using R programming

- Animate Graphs in R: Make Gorgeous Animated Plots with gganimate
- How to Make Beautiful Tables in R

Additional Resources

The resources below are provided as a mixture of optional background reading, technical reference, and useful tutorials.

- YouTube - Beautiful reports and presentations with Quarto
- How to style your Quarto docs without knowing HTML & CSS
- OPML Options